

Assignment – 3

1.enumerate()

enumerate() is a built-in Python function that adds a counter (index) to an iterable (like list, tuple, string) and returns it as an enumerate object.

Syntax: enumerate(iterable, start=0)

iterable → list, tuple, string, etc.

start → optional starting index (default is 0)

Example:

```
fruits = ["apple", "banana", "mango"]
```

```
for index, value in enumerate(fruits):
```

```
    print(index, value)
```

Output:

```
0 apple
```

```
1 banana
```

```
2 mango
```

2. isinstance()

isinstance() is used to check whether an object belongs to a particular class (data type).

Syntax:

```
isinstance(object, class)
```

Example: x = 10

```
print(isinstance(x, int))
```

Output: True

3. gc.collect()

gc.collect() is used to manually trigger Python's Garbage Collector to free unused memory. Python automatically manages memory, but sometimes we force garbage collection using this function.

Step 1: Import gc module:

```
import gc
```

Step 2: Use gc.collect():

```
import gc
```

```
collected = gc.collect()
```

```
print("Garbage collected objects:", collected)
```

Output:

It prints the number of unreachable objects that were removed from memory.